

Fourier Transform

Why?

- * Complex signals break into simple signal component, whose Analysis is very simple
- * very much useful in signal Processing, Image Processing, image Analysis & filtering
- * It is a mathematical tool used for freq Analysis of signal.
- * It works on Aperiodic signals Generally
- * FT $\begin{cases} \text{Energy signal} \rightarrow \text{finite} \\ \text{Power signal} \rightarrow \cdot \\ \text{Impulse signal} \rightarrow \delta(t) \end{cases}$

$$* f(t) \xrightarrow[\text{Time domain signal}]{\text{Fourier T/f}} \begin{matrix} F(j\omega) \\ \text{or} \\ F(\omega) \end{matrix}$$

$$F(j\omega) \xrightarrow[\text{Freq domain signal}]{\text{Inverse Fourier Transform}} f(t)$$

$$\begin{matrix} f(t) \xrightarrow{\text{FT}} F(\omega) \\ F(\omega) \xrightarrow{\text{IFT}} f(t) \end{matrix}$$

Important formula

Let $f(t)$ time domain signal
 $F(j\omega)$ freq " "

$$\left. \begin{aligned} F(j\omega) &= \int_{-\infty}^{\infty} f(t) \cdot e^{-j\omega t} dt \\ f(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega) e^{j\omega t} d\omega \end{aligned} \right\} \begin{array}{l} \text{FT} \\ \text{Pair} \end{array}$$

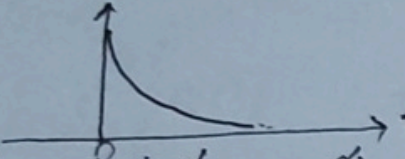
Note: $\rightarrow$ To find FT, signal has to be absolutely integrable
 $\int_{-\infty}^{\infty} f(t) \cdot dt \rightarrow \text{finite value}$

* FT is a unique.

It means no 2 signals have same Fourier Transform.

Questions on Fourier Transform

① $f(t) = e^{-at} u(t)$
find FT of $f(t)$



$$F(\omega) = \int_{-\infty}^{\infty} f(t) \cdot e^{-j\omega t} dt = \int_0^{\infty} e^{-at} \cdot e^{-j\omega t} dt$$

$$F(\omega) = \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$F(\omega) = \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty}$$

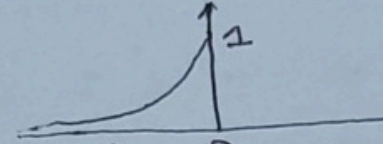
$$F(\omega) = - \left[\frac{e^{-(a+j\omega)\infty}}{a+j\omega} - \frac{e^{-(a+j\omega)0}}{a+j\omega} \right]$$

$$F(\omega) = \frac{1}{a+j\omega}$$

$$F(\omega) = \frac{1}{a+j\omega} \times \frac{a-j\omega}{a-j\omega} = \frac{a-j\omega}{a^2+\omega^2}$$

$$F(\omega) = \frac{a}{a^2+\omega^2} - \frac{j\omega}{a^2+\omega^2}$$

② $f(t) = e^{at} u(-t)$



$$F(\omega) = \int_{-\infty}^0 e^{at} \cdot e^{-j\omega t} dt$$

$$F(\omega) = \int_{-\infty}^0 e^{(a-j\omega)t} dt$$

$$F(\omega) = \left[\frac{e^{(a-j\omega)t}}{a-j\omega} \right]_{-\infty}^0$$

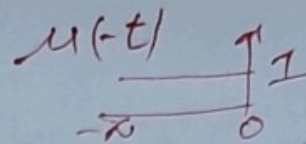
$$F(\omega) = \left[\frac{e^{(a-j\omega)0}}{a-j\omega} - \frac{e^{(a-j\omega)(-\infty)}}{a-j\omega} \right]$$

$$F(\omega) = \frac{1}{a-j\omega}$$

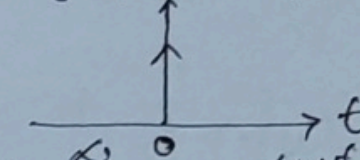
$$F(\omega) = \frac{1}{a-j\omega} \times \frac{a+j\omega}{a+j\omega}$$

$$F(\omega) = \frac{a+j\omega}{a^2+\omega^2}$$

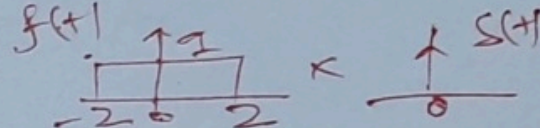
$$F(\omega) = \frac{a}{a^2+\omega^2} + \frac{j\omega}{a^2+\omega^2}$$



③ $f(t) = \delta(t)$



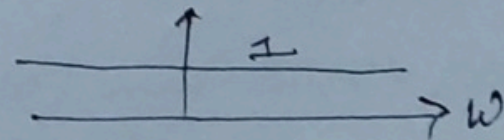
$$F(\omega) = \int_{-\infty}^{\infty} \delta(t) \cdot e^{-j\omega t} dt$$



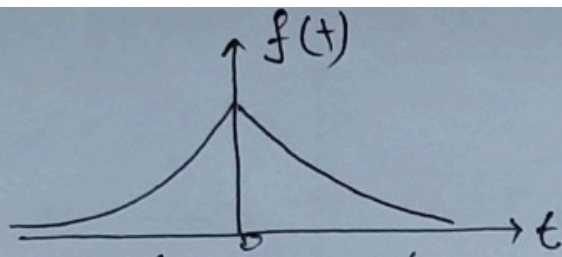
$$F(\omega) = \int_{-\infty}^{\infty} \delta(t) \cdot e^{-j\omega t} dt$$

$$= 1 \times \int_{-\infty}^{\infty} \delta(t) \cdot dt$$

$$F(\omega) = 1$$



④



$$f(t) = e^{at} u(-t) + e^{-at} u(t)$$

$$f(t) = e^{-at} u(t) \quad t \geq 0$$

$$e^{at} u(-t) \quad t \leq 0$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) \cdot e^{-j\omega t} dt$$

$$\begin{aligned} -\infty \text{ to } 0 & \quad f(t) = e^{at} \\ 0 \text{ to } \infty & \quad f(t) = e^{-at} \end{aligned}$$

$$F(\omega) = \int_{-\infty}^0 e^{at} \cdot e^{-j\omega t} dt + \int_0^{\infty} e^{-at} \cdot e^{-j\omega t} dt$$

$$F(\omega) = \int_{-\infty}^0 e^{(a-j\omega)t} dt + \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$F(\omega) = \left[\frac{e^{(a-j\omega)t}}{a-j\omega} \right]_{-\infty}^0 + \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty}$$

$$F(\omega) = \left(\frac{e^0}{a-j\omega} - \frac{e^{-\infty}}{a-j\omega} \right) - \left(\frac{e^{-\infty}}{a+j\omega} - \frac{e^0}{a+j\omega} \right)$$

$$F(\omega) = \frac{1}{a-j\omega} - 0 - 0 + \frac{1}{a+j\omega}$$

$$F(\omega) = \frac{1}{a-j\omega} + \frac{1}{a+j\omega}$$

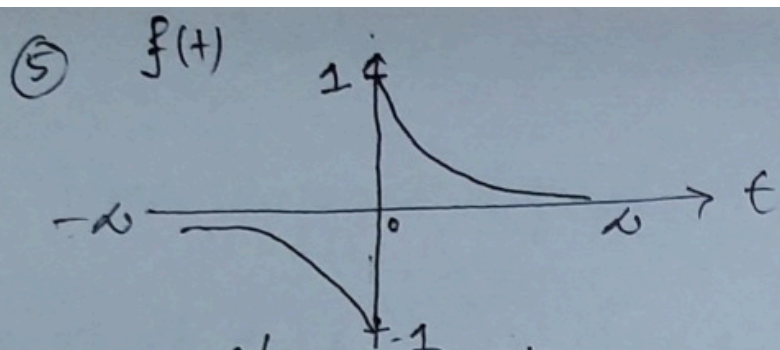
$$F(\omega) = \frac{a+j\omega + a-j\omega}{a^2 + \omega^2}$$

$$F(\omega) = \frac{2a}{a^2 + \omega^2} \quad \text{xxx}$$

$$F(-\omega) = \frac{2a}{a^2 + (-\omega)^2} = \frac{2a}{a^2 + \omega^2}$$

$$F(\omega) = F(-\omega) \rightarrow \text{even, Real}$$

$$f(t) = f(-t) \rightarrow \text{even}$$



$$f(t) = e^{-at}u(t) - e^{at}u(-t)$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) \cdot e^{-j\omega t} \cdot dt$$

$-\infty$ to ∞ $f(t)$
 $\Downarrow$ $a(t)$
 $-\infty$ to $0 \Rightarrow -e^{at}$
 0 to $\infty \Rightarrow e^{-at}$

$$F(\omega) = \int_{-\infty}^0 -e^{at} \cdot e^{-j\omega t} \cdot dt + \int_0^{\infty} e^{-at} \cdot e^{-j\omega t} \cdot dt$$

$$F(\omega) = \int_0^{\infty} e^{-(a+j\omega)t} \cdot dt - \int_{-\infty}^0 e^{(a-j\omega)t} \cdot dt$$

$$\rightarrow F(\omega) = \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty} - \left[\frac{e^{(a-j\omega)t}}{a-j\omega} \right]_{-\infty}^0$$

$$F(\omega) = \left[\frac{e^{-\infty}}{-(a+j\omega)} - \frac{e^0}{-(a+j\omega)} \right] - \left[\frac{e^0}{a-j\omega} - \frac{e^{-\infty}}{a-j\omega} \right]$$

$$F(\omega) = \frac{1}{a+j\omega} - \frac{1}{a-j\omega}$$

$$F(\omega) = \frac{a-j\omega - (a+j\omega)}{a^2 + \omega^2} = \frac{a-j\omega - a-j\omega}{a^2 + \omega^2}$$

$$F(\omega) = \frac{-2j\omega}{a^2 + \omega^2} \quad \text{xxx}$$

$$F(\omega) = \frac{2j\omega}{a^2 + (-\omega)^2} = \frac{2j\omega}{a^2 + \omega^2}$$

$$F(\omega) = -F(-\omega) \quad \text{odd}$$

Important conclusion

Time Domain signal
 $f(t)$

Freq Domain Signal
 $F(\omega)$

Real & Non-symmetrical

Complex nature $F(\omega) = \overline{F(-\omega)}$
Even Conjugate, Conjugate Symmetry

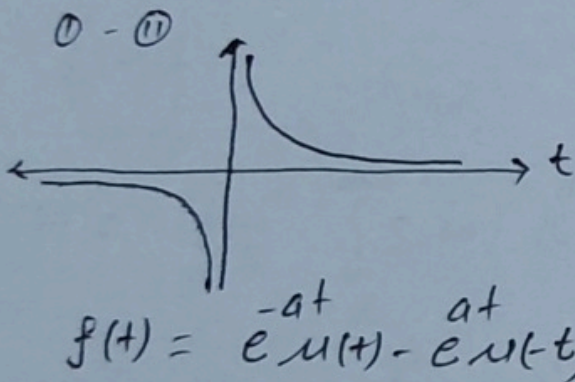
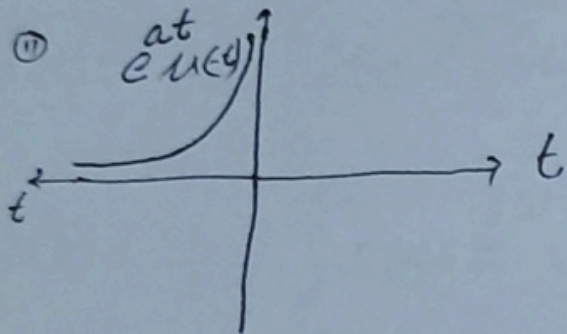
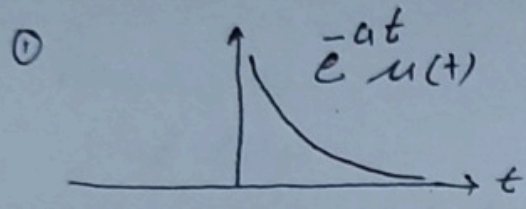
Real & Even

Real & Even ✓

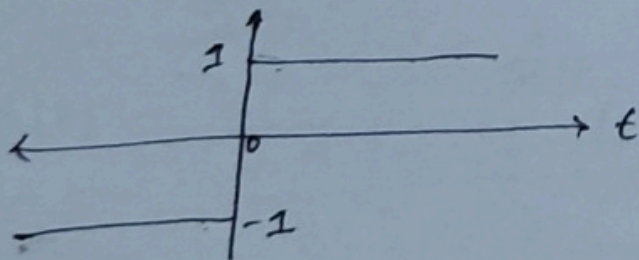
Real & odd

Imaginary & odd ✓

Special Fourier Transform



$$\lim_{a \rightarrow 0} f(t) = u(t) - u(-t) = \text{sgn}(t)$$



find FT of sgn(t) signal

$$f(t) = e^{-at} u(t) - e^{at} u(-t)$$

$$\lim_{a \rightarrow 0} f(t) = u(t) - u(-t)$$

$$F(\omega) = \frac{-2j\omega}{a^2 + \omega^2}$$

$$\lim_{a \rightarrow 0} F(\omega) = \frac{-2j\omega}{0 + \omega^2}$$

$$F(\omega) = \frac{-2j}{\omega} \times \frac{j}{j}$$

$$F(\omega) = \frac{2}{j\omega}$$

limited Fourier T.F.

$$\int_{-\infty}^{\infty} |f(t)| dt < \infty$$

$$\int_{-\infty}^{\infty} 1 dt$$

$$|t|_{-\infty}^{\infty} = \infty - (-\infty) = \infty$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$f(t)$$

$$0 - \infty = u(t) = 1$$

$$-\infty - 0 = u(-t) = -1$$

$$F(\omega) = \int_{-\infty}^0 -1 \times e^{-j\omega t} dt + \int_0^{\infty} 1 \times e^{-j\omega t} dt$$

$$= - \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-\infty}^0 + \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^{\infty}$$

$$= + \left[\frac{1}{j\omega} - \frac{e^{\infty}}{j\omega} \right] + \left[0 - \frac{1}{j\omega} \right]$$

$$= \frac{1}{j\omega} - \frac{e^{\infty}}{j\omega} + \frac{1}{j\omega}$$

$$= \frac{2}{j\omega} - \frac{e^{\infty}}{j\omega}$$

$$= \frac{2}{j\omega} - \infty = F(\omega)$$

not feasible

X

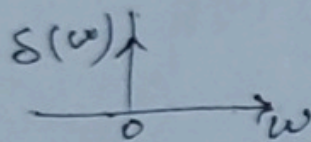
Problem Inverse Fourier Transform

i) $F(\omega) = \delta(\omega)$

find $f(t)$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) e^{j\omega t} d\omega$$



$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) e^{j(0)t} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) d\omega$$

$$f(t) = \frac{1}{2\pi} \times 1$$

$$\boxed{f(t) = \frac{1}{2\pi}}$$

$$\frac{1}{2\pi} \xrightarrow{FT} \delta(\omega)$$

$$\boxed{1 \xrightarrow{FT} 2\pi \delta(\omega)}$$

ii) $F(\omega) = 2\pi \delta(\omega - \omega_0)$

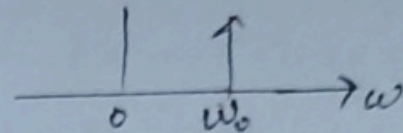
find $f(t)$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi \delta(\omega - \omega_0) e^{j\omega t} d\omega$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi \delta(\omega - \omega_0) e^{j\omega_0 t} d\omega$$

$$f(t) = e^{j\omega_0 t} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) d\omega$$

$$\boxed{f(t) = e^{j\omega_0 t}}$$



$$e^{j\omega_0 t} \xrightarrow{FT} 2\pi \delta(\omega - \omega_0)$$

$$e^{-j\omega_0 t} \xrightarrow{FT} 2\pi \delta(\omega + \omega_0)$$

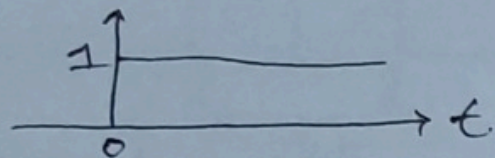
$$\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \longrightarrow \frac{2\pi}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$\cos \omega_0 t \xrightarrow{FT} \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

Special Fourier Transform

find FT of $u(t)$

$$f(t) = u(t)$$



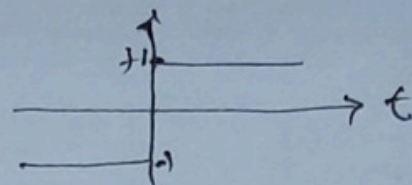
$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \quad \times$$

$$\int_{-\infty}^{\infty} |f(t) e^{-j\omega t}| dt < \infty$$

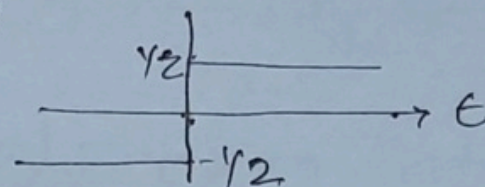
$$\int_{-\infty}^{\infty} 1 dt = |t|_0^{\infty} = \infty$$

not absolutely integrable

$$\text{sgn}(t)$$



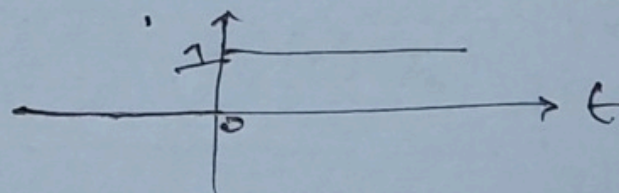
$$\frac{1}{2} \text{sgn}(t)$$



$$\frac{1}{2}$$



$$\frac{1}{2} + \frac{1}{2} \text{sgn}(t) =$$



$u(t)$

$$u(t) = \frac{1}{2} + \frac{1}{2} \text{sgn}(t)$$

$$\text{FT} \downarrow$$

$$\pi \delta(\omega)$$

$$\text{FT} \downarrow$$

$$\frac{1}{j\omega}$$

$$u(t) \xrightarrow{\text{FT}} \pi \delta(\omega) + \frac{1}{j\omega}$$

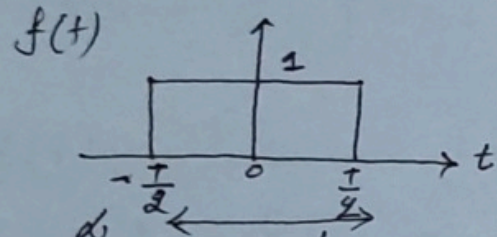
limited Fourier Transform

FT $\rightarrow$ Energy

FT $\rightarrow$ Power
limited

Fourier Transform of Rectangular Pulse

Find Fourier Transform of



$$\text{Area} = T \times 1 = T$$

$$f(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$= \int_{-T/2}^{T/2} 1 \times e^{-j\omega t} dt = \int_{-T/2}^{T/2} e^{-j\omega t} dt$$

$$f(\omega) = \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-T/2}^{T/2} = \frac{e^{-j\omega T/2}}{-j\omega} + \frac{e^{j\omega T/2}}{j\omega}$$

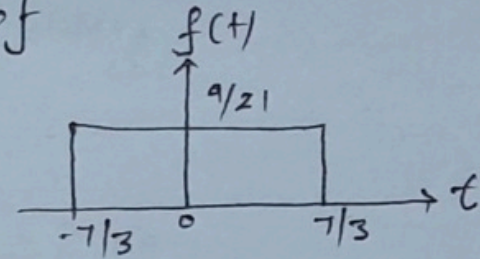
$$f(\omega) = \frac{e^{j\omega T/2}}{j\omega} - \frac{e^{-j\omega T/2}}{j\omega}$$

$$= \frac{2}{\omega} \left[\frac{e^{j\omega T/2} - e^{-j\omega T/2}}{j2} \right]$$

$$f(\omega) = \frac{2}{\omega} \sin(\omega T/2) \quad \text{xxx}$$

$$f(\omega) = \frac{2}{\omega} \frac{\sin(\omega T/2)}{\omega T/2} \times \omega T/2$$

find FT of



$$f(\omega) = \text{Area} \text{Sa}(\omega \times \frac{\text{width}}{2})$$

$$= \frac{1}{3} \times \frac{1}{2} \text{Sa}(\omega \times \frac{T}{3})$$

$$f(\omega) = \frac{2}{3} \text{Sa}(\omega T/3) \quad \text{xxx}$$

$$\frac{\sin t}{t} = \text{Sa}(t)$$

$$= \frac{2}{\omega} \times \frac{\omega T/2}{2} \text{Sa}(\omega T/2)$$

$$= \boxed{T \text{Sa}(\omega T/2)} \quad \text{v.v Imp}$$

→ Area

Properties of Fourier Transform

① Time shifting

$$f(t) \xrightarrow{FT} F(\omega)$$

$$f(t-t_0) \xrightarrow{FT} ?$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$f(t-t_0) \xrightarrow{FT} ?$$

$$FT = \int_{-\infty}^{\infty} f(t-t_0) e^{-j\omega t} dt$$

$$t-t_0 = u$$

$$t = u + t_0$$

$$dt = du$$

$$FT = \int_{-\infty}^{\infty} f(u) e^{-j\omega(u+t_0)} du$$

$$FT = \int_{-\infty}^{\infty} f(u) e^{-j\omega u} e^{-j\omega t_0} du$$

$$FT = e^{-j\omega t_0} \int_{-\infty}^{\infty} f(u) e^{-j\omega u} du$$

$$FT = e^{-j\omega t_0} F(\omega)$$

$$f(t) \xrightarrow{FT} F(\omega)$$

$$f(t-t_0) \xrightarrow{FT} e^{-j\omega t_0} F(\omega)$$

$$f(t+t_0) \xrightarrow{FT} e^{j\omega t_0} F(\omega)$$

$$f(t+t_0) + f(t-t_0) \xrightarrow{FT} 2F(\omega) \left[\frac{e^{j\omega t_0} + e^{-j\omega t_0}}{2} \right]$$

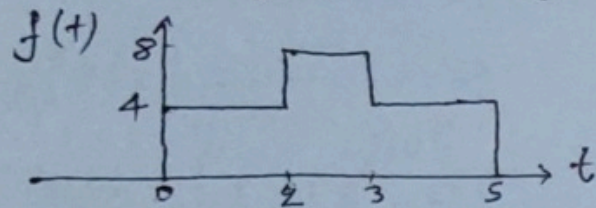
$$\xrightarrow{FT} 2F(\omega) \cos(\omega t_0)$$

$$f(t+t_0) - f(t-t_0) \xrightarrow{FT} F(\omega) \left[\frac{e^{j\omega t_0} - e^{-j\omega t_0}}{2j} \right] \times 2j$$

$$\xrightarrow{FT} 2j F(\omega) \sin(\omega t_0)$$

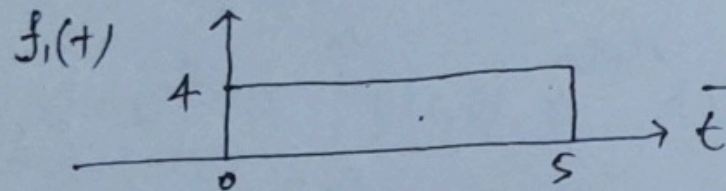
Problem on Time Shifting Properties

find F.T of given signal



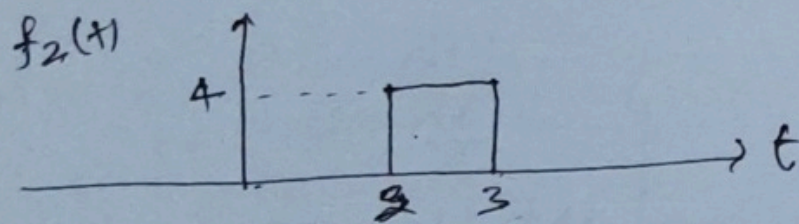
FT

$$e^{-j\omega \frac{5}{2}} \left[20 \text{sinc}\left(\frac{\omega 5}{2}\right) + 4 \text{sinc}\left(\frac{\omega}{2}\right) \right]$$



FT

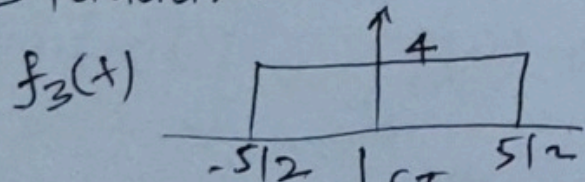
$$20 \text{sinc}\left(\frac{\omega 5}{2}\right) e^{-j\omega 5/2}$$



FT

$$4 \text{sinc}\left(\frac{\omega}{2}\right) e^{-j\omega \frac{5}{2}}$$

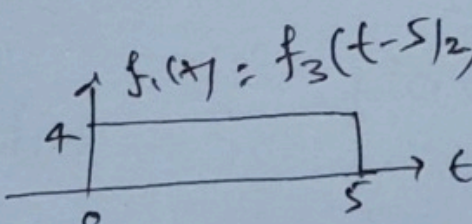
standard



FT

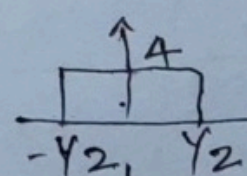
$$5 \times 4 \text{sinc}\left(\frac{\omega 5}{2}\right)$$

Time shift by 5/2

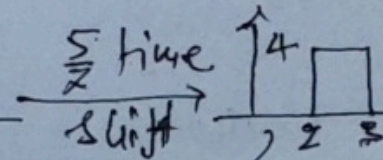


$$e^{-j\omega \frac{5}{2}} \times 20 \text{sinc}\left(\frac{\omega 5}{2}\right)$$

standard



$$1 \times 4 \text{sinc}\left(\frac{\omega}{2}\right)$$



$$4 \text{sinc}\left(\frac{\omega}{2}\right) e^{-j\omega \frac{5}{2}}$$

② Time scaling Property

$$f(t) \xrightarrow{FT} F(\omega)$$

$$f(at) \xrightarrow{FT} ?$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$F[f(at)] = \int_{-\infty}^{\infty} f(at) e^{-j\omega t} dt$$

$$\left. \begin{aligned} at &= u \\ t &= \frac{u}{a} \\ dt &= \frac{1}{a} du \end{aligned} \right\}$$

$$F[f(at)] = \int_{-\infty}^{\infty} f(u) e^{-j\omega \frac{u}{a}} \cdot \frac{1}{a} du$$

$$= \frac{1}{a} \int_{-\infty}^{\infty} f(u) e^{-j(\frac{\omega}{a})u} du$$

$$F[f(at)] = \frac{1}{a} F\left(\frac{\omega}{a}\right)$$

$$f(t) \xrightarrow{FT} F(\omega)$$

$$f(at) \xrightarrow{FT} \frac{1}{a} F(\omega/a)$$

case 1, $a > 0$

$$f(at) \xrightarrow{FT} \frac{1}{a} F(\omega/a)$$

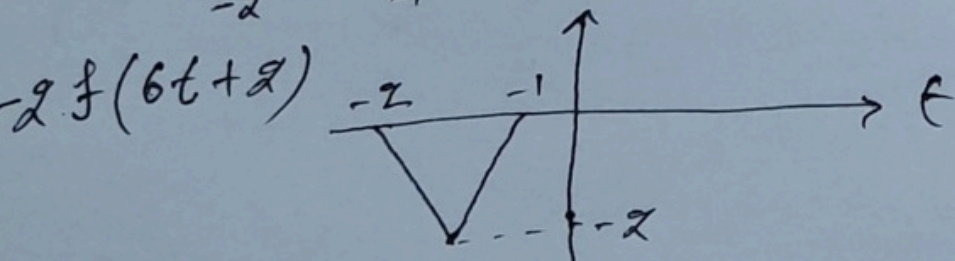
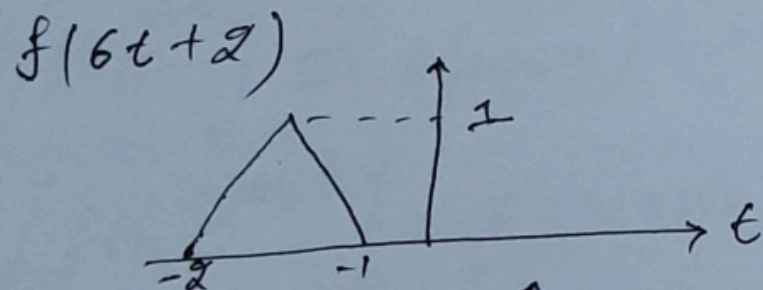
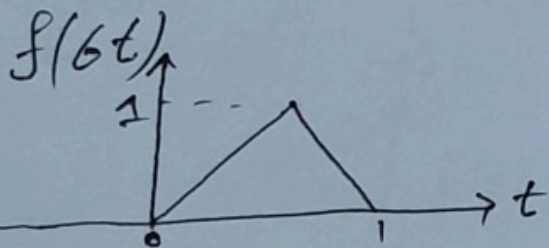
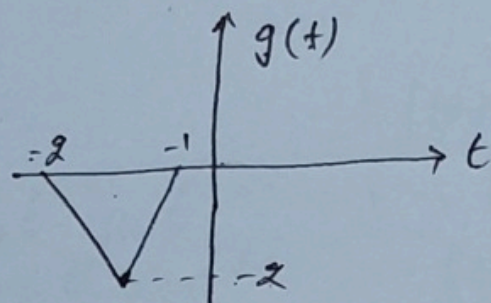
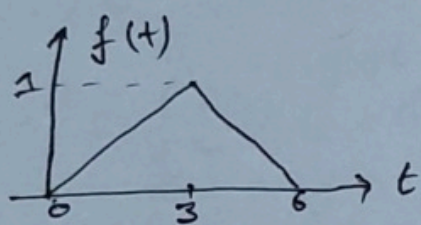
case 2, $a < 0$

$$f(at) \xrightarrow{FT} -\frac{1}{a} F(\omega/a)$$

$$f(at) \xrightarrow{FT} \boxed{\frac{1}{|a|} F(\omega/a)}$$

Problem on Time scaling

- ① Fourier Transform of $f(t) = F(\omega)$
find Fourier Transform of $g(t)$ in term of $F(\omega)$.



$$f(t) \xrightarrow{FT} F(\omega)$$

$$f(6t) \xrightarrow{FT} \frac{1}{6} F\left(\frac{\omega}{6}\right)$$

$$f(6t+2) \xrightarrow{FT} \frac{1}{6} F\left(\frac{\omega}{6}\right) e^{j\omega 2}$$

$$-2f(6t+2) \xrightarrow{FT} -\frac{2}{6} F\left(\frac{\omega}{6}\right) e^{j\omega 2}$$

$$= \underline{\underline{-\frac{1}{3} F\left(\frac{\omega}{6}\right) e^{j\omega 2}}}$$

③ Frequency Shifting Property

$$f(t) \xrightarrow{FT} F(\omega)$$

$$f(t) e^{j\omega_0 t} \xrightarrow{FT} ?$$

$$f(t) e^{-j\omega_0 t} \xrightarrow{FT} ?$$

$$F(\omega) = \int_{-\infty}^{\infty} \underline{f(t)} \cdot \underline{e^{-j\omega t}} \cdot dt$$

$$F[f(t) e^{j\omega_0 t}] = \int_{-\infty}^{\infty} f(t) e^{j\omega_0 t} \cdot e^{-j\omega t} \cdot dt$$

$$= \int_{-\infty}^{\infty} f(t) e^{-j(\omega - \omega_0)t} \cdot dt$$

$$F[f(t) e^{j\omega_0 t}] = F(\omega - \omega_0)$$

$$F[f(t) e^{-j\omega_0 t}] = F(\omega + \omega_0)$$

$$f(t) \xrightarrow{FT} F(\omega)$$

$$f(t) e^{j\omega_0 t} \xrightarrow{FT} F(\omega - \omega_0)$$

$$f(t) e^{-j\omega_0 t} \xrightarrow{FT} F(\omega + \omega_0)$$

$$f(t) \left[\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2} \right] \xrightarrow{FT} \frac{F(\omega - \omega_0) + F(\omega + \omega_0)}{2}$$

$$f(t) \cos \omega_0 t \xrightarrow{FT} \frac{1}{2} [F(\omega - \omega_0) + F(\omega + \omega_0)]$$

Imp**

$$f(t) \left[\frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} \right] \xrightarrow{FT} \frac{F(\omega - \omega_0) - F(\omega + \omega_0)}{2j}$$

$$f(t) \sin \omega_0 t \xrightarrow{FT} \frac{1}{2j} [F(\omega - \omega_0) - F(\omega + \omega_0)]$$

Imp**

$$f(t) \xrightarrow{FT} \checkmark$$

$$\begin{array}{c} \swarrow \quad \searrow \\ \cos \quad \sin \end{array} \xrightarrow{FT} ?$$

Problem based freq shifting Property

① find Fourier Transform of Given

signal: $e^{-10|t|} \sin 100t$

$$f(t) = e^{-10|t|} \xrightarrow{FT} \frac{2a}{a^2 + \omega^2} = F(\omega)$$

$$\xrightarrow{FT} \frac{20}{100 + \omega^2}$$

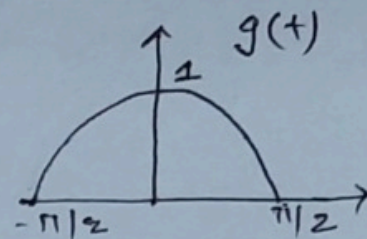
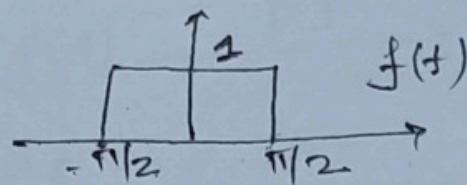
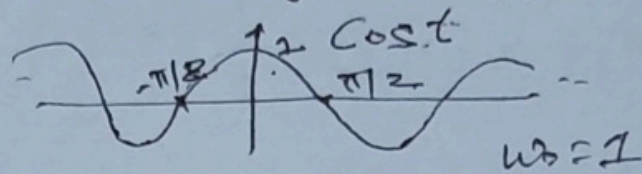
$$f(t) \sin \omega_0 t \xrightarrow{FT} \frac{1}{2j} [F(\omega - \omega_0) - F(\omega + \omega_0)]$$

$\omega_0 = 100$

$$e^{-10|t|} \sin 100t \xrightarrow{FT} \frac{1}{2j} \left[\frac{20}{100 + (\omega - 100)^2} - \frac{20}{100 + (\omega + 100)^2} \right]$$

Ans xx

② find FT of $g(t)$ signal



$$f(t) \xrightarrow{FT} \pi \text{Sa}\left(\omega \frac{\pi}{2}\right) = F(\omega)$$

$$f(t) \cos t \xrightarrow{FT} \frac{1}{2} [F(\omega - \omega_0) + F(\omega + \omega_0)]$$

$$f(t) \cos t \xrightarrow{FT} \frac{1}{2} \left[\pi \text{Sa}\left((\omega - 1) \frac{\pi}{2}\right) + \pi \text{Sa}\left((\omega + 1) \frac{\pi}{2}\right) \right]$$

Ans xx

④ Time Differentiation Property

$$f(t) \xrightarrow{FT} F(\omega)$$

$$\frac{d}{dt} f(t) \xrightarrow{FT} ?$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$F(\omega) \rightarrow \text{Given}$$

$$f(t) \rightarrow ?$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

Diff. w.r.t time

$$\frac{df}{dt} = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} (j\omega) d\omega$$

$$\frac{df}{dt} = (j\omega) \underbrace{\frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega}_{f(t)}$$

$$\frac{df}{dt} = (j\omega) \mathbf{F}^{-1}(F(\omega))$$

taking FT on both sides

$$F\left\{\frac{df}{dt}\right\} = (j\omega) \times F(\omega)$$

$$f(t) \xrightarrow{FT} F(\omega)$$

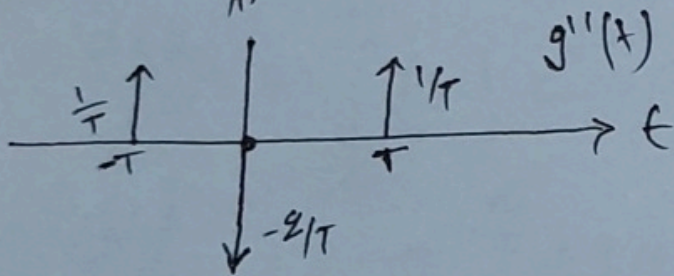
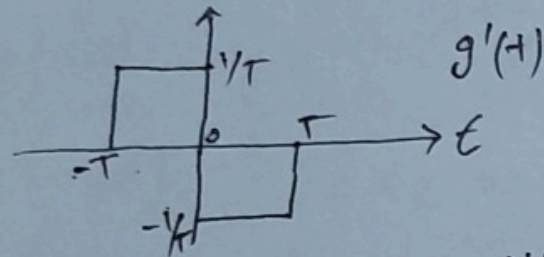
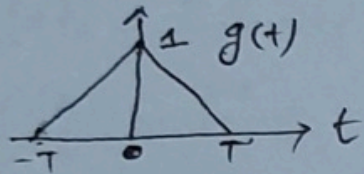
$$\frac{df}{dt} \xrightarrow{FT} \boxed{j\omega F(\omega)} \quad \times \times \times$$

$$\frac{d^n f}{dt^n} \xrightarrow{FT} \boxed{(j\omega)^n F(\omega)} \quad \times \times \times$$

$$f(t) \xrightarrow{FT} F(\omega) \xrightarrow{IFT} f(t)$$

Problem Based on Time Diff. Property

① find Fourier Transform of $g(t)$ signal.



$$g''(t) = \frac{1}{T} \delta(t+T) - \frac{2}{T} \delta(t) + \frac{1}{T} \delta(t-T)$$

$$\begin{aligned} \downarrow \text{FT} \\ G''(\omega) &= \frac{1}{T} e^{j\omega T} - \frac{2}{T} \times 1 + \frac{1}{T} e^{-j\omega T} \\ &= -\frac{2}{T} + \frac{1}{T} \left[\frac{e^{j\omega T} + e^{-j\omega T}}{2} \right] \times 2 \end{aligned}$$

$$s(t) \xrightarrow{\frac{d}{dt}} u(t) \xrightarrow{\frac{d}{dt}} v(t) \xrightarrow{\frac{d}{dt}} p(t)$$

We know

$$s(t) \xrightarrow{\text{FT}} 1$$

$$\begin{aligned} s(t+T) &\rightarrow 1 \times e^{j\omega T} \\ \frac{1}{T} s(t-T) &\rightarrow \frac{1}{T} \times e^{-j\omega T} \end{aligned}$$

$$\begin{aligned} &-\frac{2}{T} + \frac{2}{T} \cos \omega T \\ &= -\frac{2}{T} [1 - \cos \omega T] \\ &= -\frac{4}{T} \sin^2 \left(\frac{\omega T}{2} \right) \end{aligned}$$

$$f(t) \rightarrow f(\omega)$$

$$f''(t) \rightarrow (j\omega)^2 \times f(\omega)$$

$$g(t) \xrightarrow{\text{FT}} +\frac{4}{T} \frac{\sin^2 \left(\frac{\omega T}{2} \right)}{\omega^2}$$

$$\begin{aligned} &= \frac{4 \sin^2 \left(\frac{\omega T}{2} \right) \times T}{\omega^2 \times \frac{\pi T}{4} \times 4} \\ &= 4 \frac{\sin^2 \left(\frac{\omega T}{2} \right) \times T}{\left(\frac{\omega T}{2} \right)^2 \times 4} \end{aligned}$$

$$= T \text{Sa}^2 \left(\frac{\omega T}{2} \right)$$

$$1 - \cos 2\phi = 2 \sin^2 \left(\frac{2\phi}{2} \right) = \text{Area Sa}^2 \left(\omega \times \frac{\text{width}}{2} \right)$$

⑤ Frequency Diff. Property

$$f(t) \xrightarrow{FT} F(\omega)$$

$$t f(t) \xrightarrow{FT} ?$$

$$F(\omega) = \int_{-\infty}^{\infty} \underline{f(t)} \cdot \underline{e^{-j\omega t}} \cdot dt$$

Diff this eq w.r.t ω

$$\frac{dF(\omega)}{d\omega} = \int_{-\infty}^{\infty} f(t) \cdot e^{-j\omega t} (-jt) \cdot dt$$

$$\frac{dF}{d\omega} = -j \int_{-\infty}^{\infty} t f(t) \cdot e^{-j\omega t} \cdot dt$$

$$\frac{1}{-j} \frac{dF(\omega)}{d\omega} = \int_{-\infty}^{\infty} t f(t) e^{-j\omega t} \cdot dt$$

$$\frac{-1}{j} \times \frac{j}{j} \frac{dF(\omega)}{d\omega} = \int_{-\infty}^{\infty} t \cdot f(t) \cdot e^{-j\omega t} \cdot dt$$

$$j \frac{dF(\omega)}{d\omega} = \int_{-\infty}^{\infty} \underline{t f(t)} \cdot \underline{e^{-j\omega t}} \cdot dt$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) \cdot \underline{e^{-j\omega t}} \cdot dt$$

$$f(t) \xrightarrow{FT} F(\omega)$$

$$t f(t) \xrightarrow{FT} \boxed{j \frac{dF(\omega)}{d\omega}} \quad \text{xxx}$$

$$f(t) \xrightarrow{FT} \checkmark$$

$$t f(t) \xrightarrow{FT} ?$$

Problem based on Freq. Diff. Property

① $f(t) = t e^{-at} u(t)$

find Fourier Transform

$$f(t) \xrightarrow{FT} F(\omega)$$

$$t f(t) \xrightarrow{FT} j \frac{dF}{d\omega}$$

$$e^{-at} u(t) \xrightarrow{FT} \frac{1}{a + j\omega}$$

$$t e^{-at} u(t) \xrightarrow{FT} j \frac{d}{d\omega} \left[\frac{1}{a + j\omega} \right]$$

$$\begin{aligned} \frac{d}{d\omega} \left[\frac{1}{a + j\omega} \right] &= \frac{(a + j\omega)(0) - 1 \times j}{(a + j\omega)^2} \\ &= \frac{-j}{(a + j\omega)^2} \end{aligned}$$

$$t e^{-at} u(t) \xrightarrow{FT} j \times \left[\frac{-j}{(a + j\omega)^2} \right]$$

$$= \boxed{\frac{1}{(a + j\omega)^2}} \text{xxx}$$

$$t^n e^{-at} u(t) \xrightarrow{FT} \frac{n!}{(a + j\omega)^{n+1}}$$

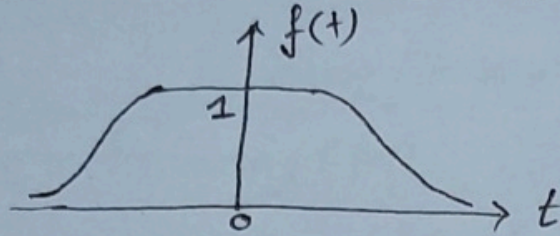
$$3! = 3 \times 2 \times 1 = 6$$

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

Fourier Transform of Gaussian Pulse

$$f(t) = e^{-\pi t^2}$$

$$f(t) \xrightarrow{FT} F(\omega)$$



$$f(t) \xrightarrow{FT} F(\omega)$$

$$t f(t) \xrightarrow{FT} j \frac{dF}{d\omega}$$

* Gaussian Pulse $\xrightarrow{FT}$ Gaussian Pulse

$$\frac{df}{dt} = e^{-\pi t^2} \times (-2t\pi)$$

$$\frac{df}{dt} = -2\pi t e^{-\pi t^2} = (-2\pi) \underline{t e^{-\pi t^2}} = (-2\pi) t f(t)$$

$$f(t) \xrightarrow{FT} F(\omega)$$

$$\frac{df}{dt} \longrightarrow j\omega F(\omega)$$

$$f(t) \longrightarrow F(\omega)$$

$$t f(t) \longrightarrow j \frac{dF}{d\omega}$$

$$-2\pi t f(t) \xrightarrow{FT} -2\pi j \frac{dF}{d\omega}$$

$$j\omega F(\omega) = -2\pi j \frac{dF}{d\omega}$$

$$\frac{j\omega F(\omega)}{-2\pi j} = \frac{dF}{d\omega}$$

$$\frac{\omega F(\omega)}{-2\pi} = \frac{dF}{d\omega}$$

$$-\frac{\omega}{2\pi} d\omega = \frac{dF(\omega)}{F(\omega)}$$

Integrating on both sides

$$\log_e F(\omega) = \frac{-1}{2\pi} \frac{\omega^2}{2} + C$$

$$e^{-\frac{\omega^2}{4\pi} + C} = F(\omega)$$

$$F(\omega) = e^{-\frac{\omega^2}{4\pi}} \cdot e^C \quad \text{let } C=0$$

$$F(\omega) = e^{-\omega^2/4\pi} \quad \text{xxx}$$

Proofⁿ
 $\log_e x = z + C$
 $e^{z+C} = x$

Duality of Fourier Transform

Time Domain

Freq Domain

$$f(t) \xrightarrow{FT} F(\omega)$$

$$F(t) \xrightarrow{FT} 2\pi f(-\omega)$$

Solved Problem

① find FT of $f(t) = \frac{1}{a+jt}$

$$e^{-at} u(t) \xrightarrow{FT} \frac{1}{a+j\omega}$$

$$\frac{1}{a+jt} \xrightarrow{FT} 2\pi e^{-a(\omega)} u(-\omega)$$

$$= \boxed{2\pi e^{-a\omega} u(-\omega)}$$

②

$f(t) = \frac{1}{\pi t}$, find FT

$2/j\omega$

$$\text{sgn}(t) \xrightarrow{FT} \frac{2}{j\omega}$$

$$\frac{2}{j\omega} \xrightarrow{FT} 2\pi \text{sgn}(-\omega)$$

$$\frac{1}{j\omega} \xrightarrow{FT} \pi \text{sgn}(-\omega)$$

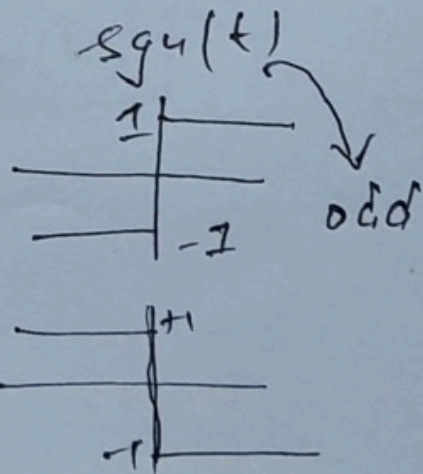
multiply j/π

$$\frac{1}{\pi t} \xrightarrow{FT} j \cancel{\pi} \text{sgn}(-\omega)$$

$$= j \text{sgn}(-\omega)$$

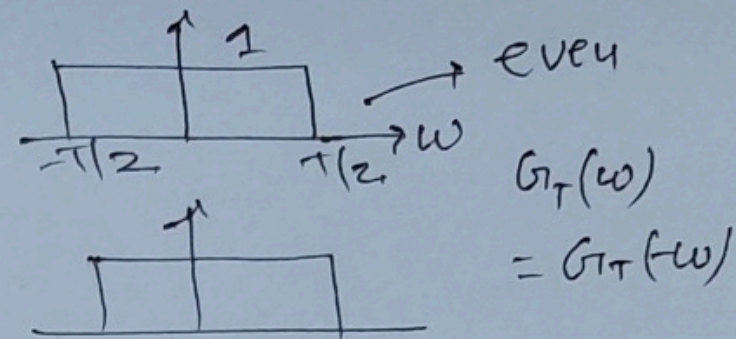
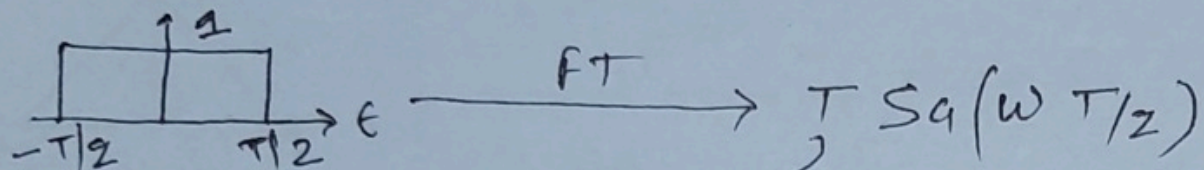
$$\Rightarrow \text{sgn}(-\omega) = -\text{sgn}(\omega)$$

$$\boxed{-j \text{sgn}(\omega)}$$



Problem based on Duality Property

③ find FT of $f(t) = \text{Sa}(t) = \frac{\sin t}{t}$



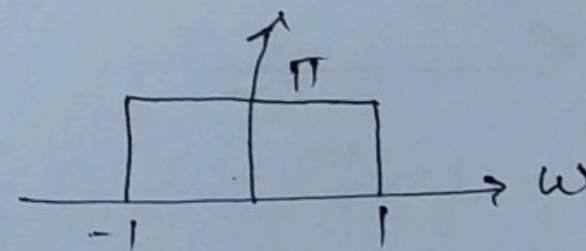
$$T \text{Sa}\left(t \frac{T}{2}\right) \xrightarrow{\text{FT}} 2\pi G_T(-w)$$

$$T \text{Sa}\left(t \frac{T}{2}\right) \xrightarrow{\text{FT}} 2\pi G_T(w)$$

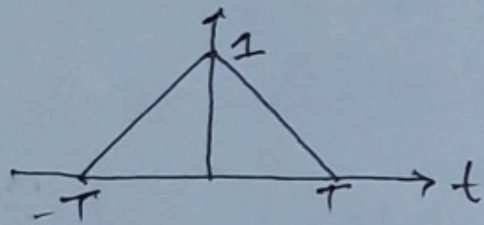
$$\frac{T}{2} = 1 \Rightarrow T = 2$$

$$2 \text{Sa}(t) \xrightarrow{\text{FT}} 2\pi G_2(w)$$

$$\text{Sa}(t) \xrightarrow{\text{FT}} \boxed{\pi G_2(w)} \quad \times \times$$



④ find FT. of $f(t) = \text{sinc}^2(t) = \frac{\sin^2 t}{t^2}$



$$\xrightarrow{\text{FT}} \text{Area } \text{sinc}^2\left(\omega \times \frac{\text{width}}{4}\right)$$

$$\xrightarrow{\text{FT}} \frac{1}{2} \times 2T \times 1 \text{ sinc}^2\left(\omega \frac{T}{2}\right)$$

$$T \text{ sinc}^2\left(\omega T/2\right)$$

$$T \text{ sinc}^2\left(t T/2\right) \xrightarrow{\text{FT}} 2\pi \text{Tri}_{\frac{2T}{2}}(-\omega)$$

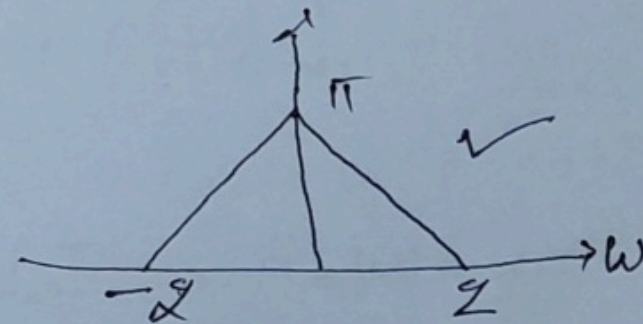
$$2\pi \text{Tri}_{\frac{2T}{2}}(\omega)$$

$$\frac{T}{2} = 1 \Rightarrow T = 2$$

$$2 \text{ sinc}^2(t) \xrightarrow{\text{FT}} 2\pi \text{Tri}_1(\omega)$$

$$\text{sinc}^2(t) \xrightarrow{\text{FT}} \boxed{\pi \text{Tri}_{\frac{1}{2}}(\omega)}$$

$$\text{Tri}_1(-\omega) = \text{Tri}_1(\omega)$$



$f(\omega)$

Time Reversal Property

$$f(t) \xrightarrow{FT} F(\omega)$$

$$f(-t) \xrightarrow{FT} F$$

$$F(\omega) = \int_{t=-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$F[f(-t)] = \int_{t=-\infty}^{\infty} f(-t) e^{-j\omega t} dt$$

$$\begin{array}{l|l} -t = u & t = -\infty \rightarrow u = \infty \\ t = -u & t = \infty \rightarrow -u = \infty \\ dt = -du & \int_{-\infty}^{\infty} \rightarrow \int_{\infty}^{-\infty} \end{array} \quad \begin{array}{l} u = \infty \\ u = -\infty \end{array}$$

$$F[f(u)] = \int_{u=\infty}^{-\infty} f(u) e^{-j\omega(-u)} (-du)$$

$$\begin{aligned} F(f(u)) &= - \int_{\infty}^{-\infty} f(u) e^{-j(-\omega)u} du \\ F[f(u)] &= \int_{-\infty}^{\infty} f(u) e^{-j(-\omega)u} du \\ &= F(-\omega) \end{aligned}$$

$$\left. \begin{array}{l} f(t) \xrightarrow{FT} F(\omega) \\ f(-t) \xrightarrow{FT} F(-\omega) \end{array} \right\}$$

Example

$$u(t) \xrightarrow{FT} \pi \delta(\omega) + \frac{1}{j\omega}$$

$$u(-t) \xrightarrow{FT} \pi \delta(-\omega) + \frac{1}{j(-\omega)}$$

$$\begin{array}{c} \uparrow 1 \\ \text{---} \rightarrow t \end{array} \quad \begin{array}{c} \uparrow 1 \\ \text{---} \rightarrow \omega \end{array} \quad \boxed{\pi \delta(\omega) - \frac{1}{j\omega}}$$

$s(\omega) = s^*(\omega)$ F.T of $u(t)$

Concept of Area in Time & Freq Domain

let $f(t)$ lies from $-\infty$ to $+\infty$

then Area of $f(t) = \int_{-\infty}^{\infty} f(t) \cdot dt$

let $F(\omega)$ lies from $-\infty$ to $+\infty$

then Area of $F(\omega) = \int_{-\infty}^{\infty} F(\omega) \cdot d\omega$

Fourier Transform

$$F(\omega) = \int_{-\infty}^{\infty} f(t) \cdot e^{-j\omega t} \cdot dt$$

put $\omega = 0$

$$F(0) = \int_{-\infty}^{\infty} f(t) \cdot e^{-j(0)t} \cdot dt$$

$$F(0) = \int_{-\infty}^{\infty} f(t) \cdot dt$$

$F(0) = \text{Area of } f(t) \text{ signal.}$

Conclusion:

$$f(t) \xrightarrow{\text{FT}} F(\omega)$$

$$\frac{\text{Area}}{F(\omega)} \Big|_{\omega=0}$$

$$\frac{\text{Area}}{2\pi f(0)}$$

Inverse Fourier Transform

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) \cdot e^{j\omega t} \cdot d\omega$$

put $t = 0$

$$f(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) \cdot e^{j\omega(0)} \cdot d\omega$$

$$f(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) \cdot d\omega$$

$$2\pi f(0) = \int_{-\infty}^{\infty} F(\omega) \cdot d\omega = \text{Area of } F(\omega)$$

Problem Based on Area Property

① Find out the Area of $Sa(t) = \frac{\sin t}{t}$

OR
Solve $\int_{-\infty}^{\infty} \frac{\sin t}{t} \cdot dt$

$$\int_{-\infty}^{\infty} f(t) \cdot dt = F(\omega) \Big|_{\omega=0}$$

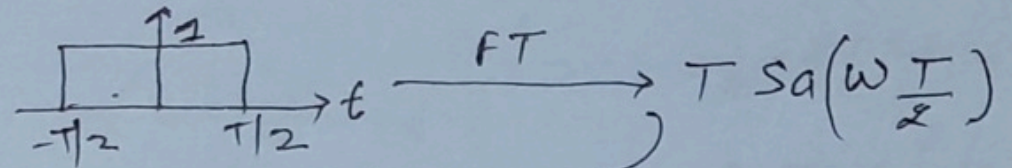
$$F(\omega) = \pi \text{ Gate}_2(\omega)$$

$$\text{at } \omega = 0$$

$$F(0) = \pi$$

$$\int_{-\infty}^{\infty} \frac{\sin t}{t} \cdot d\omega = \pi$$

xxx



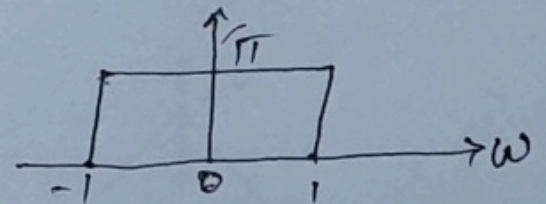
$$T \text{Sa}\left(t \frac{T}{2}\right) \xrightarrow{\text{FT}} 2\pi \times \text{Gate}_T(-\omega)$$

$$T \text{Sa}\left(t \frac{T}{2}\right) \xrightarrow{\text{FT}} 2\pi \text{Gate}_T(\omega)$$

$$\text{Put } t = T = 2$$

$$2 \text{Sa}(t) \xrightarrow{\text{FT}} 2\pi \text{Gate}_2(\omega)$$

$$\text{Sa}(t) \xrightarrow{\text{FT}} \pi \text{Gate}_2(\omega)$$



Q Find out the Area of $\underline{Sa^2(t)} = \frac{\sin^2 t}{t^2}$

Solve $\int_{-\infty}^{\infty} \frac{\sin^2 t}{t^2} dt$

Area Property

$$\int_{-\infty}^{\infty} f(t) dt = F(\omega) \Big|_{\omega=0}$$

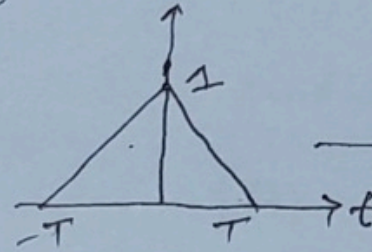
$$F(\omega) = \pi \text{Triangular}_4(\omega)$$

at $\omega=0$

$$F(0) = \pi$$

xxxx

$$\int_{-\infty}^{\infty} \frac{\sin^2 t}{t^2} dt = \pi$$



$$FT \rightarrow \frac{1}{2} \times 2T \times 1 \text{Sa}^2\left(\omega \frac{T}{2}\right)$$

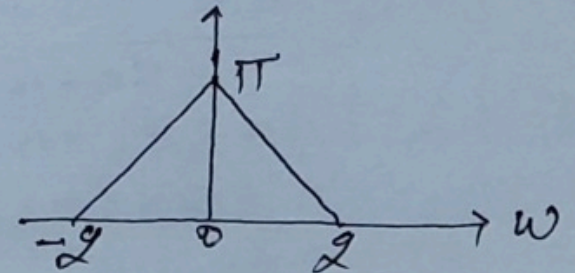
$$\rightarrow T \text{Sa}^2\left(\omega \frac{T}{2}\right)$$

$$T \text{Sa}^2\left(t \frac{T}{2}\right) \xrightarrow{FT} 2\pi \text{Triangular}_{2T}(\omega)$$

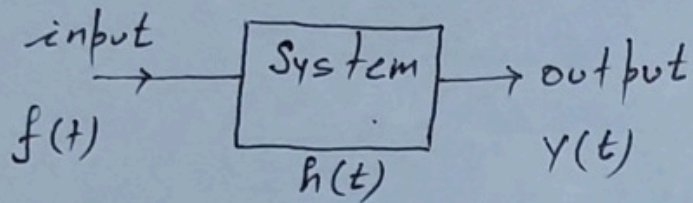
Put $T=2$

$$2 \text{Sa}^2(t) \xrightarrow{FT} 2\pi \text{Triangular}_4(\omega)$$

$$\text{Sa}^2(t) \xrightarrow{FT} \pi \text{Triangular}_4(\omega)$$



Convolution Property of Fourier Transform



Convolution:- To find out the response

In time Domain

$$y(t) = f(t) \otimes h(t)$$

$$y(t) = \int_{-\infty}^{\infty} f(\tau) h(t-\tau) d\tau$$

operation involved

- i) $f(t) \rightarrow f(\tau)$
- ii) $h(t) \rightarrow h(\tau)$
- iii) $h(\tau) \rightarrow h(-\tau+t)$ Time shift
- iv) $f(\tau) \cdot h(-\tau+t)$ multiply & find limit
- v) Integration

complex method

In Freq Domain

$$Y(\omega) = F(\omega) \cdot H(\omega)$$

$$Y(\omega) \xrightarrow{\text{IFT}} y(t)$$

$$Y(\omega) = \frac{1}{b-a} \neq \frac{1}{b-a}$$

taking inverse F.T

$$y(t) = \frac{1}{b-a} e^{-at} u(t) - \frac{1}{b-a} e^{-bt} u(t)$$

$$y(t) = \frac{1}{b-a} \left[e^{-at} - e^{-bt} \right] u(t)$$

Ans

Example

$$\textcircled{1} f(t) = e^{-at} u(t), h(t) = e^{-bt} u(t)$$

find o/p response.

$$f(t) = e^{-at} u(t) \xrightarrow{\text{FT}} \frac{1}{a+j\omega}$$

$$h(t) = e^{-bt} u(t) \xrightarrow{\text{FT}} \frac{1}{b+j\omega}$$

$$Y(\omega) = F(\omega) H(\omega)$$

$$Y(\omega) = \frac{1}{(a+j\omega)(b+j\omega)} = \frac{A}{a+j\omega} + \frac{B}{b+j\omega}$$

$$1 = A(b+j\omega) + B(a+j\omega)$$

$$1 = bA + aB + j\omega(A+B)$$

$$1 = bA + aB + j\omega(A+B)$$

$$bA + aB = 1$$

$$A + B = 0 \Rightarrow A = -B$$

$$-bB + aB = 1$$

$$B = \frac{1}{a-b} \quad \checkmark$$

$$A = \frac{1}{b-a} \quad \checkmark$$